Smart Sensor Anomaly Dashboard

High School Intern Group Project

Infiswift Technologies • Summer 2026

Interns	Sudarshini Seth, Arshita Shrivastava, Srivatsa Patnala
Start Date	June 8, 2026
Duration	4 weeks (ending July 4, 2026)
Mentor / CTO	Sivakumar Venkatesan

1. Background & Motivation

Infiswift Technologies builds enterprise AI and IoT solutions that process real-time sensor streams from industrial equipment, smart buildings, and connected devices. A recurring challenge across all of these deployments is the timely detection of anomalous sensor readings — sudden spikes, unexpected dropouts, or subtle long-term drift — before they cascade into equipment failures or costly downtime.

This project gives hands-on exposure to the full data science and software development lifecycle: ingesting and exploring time-series data, training machine learning models, exposing predictions through a REST API, and presenting results on an interactive web dashboard. The domain is directly aligned with Infiswift's core product area, so the skills and patterns interns develop here are immediately transferable to real client projects.

2. Project Objectives

By the end of four weeks, the team will have collaboratively built and demonstrated:

- A trained anomaly detection model capable of identifying sensor faults in real time.
- A FastAPI-based REST backend that serves model predictions and aggregated statistics.
- A frontend dashboard (React or plain HTML/JS) that visualises the sensor time series and highlights anomalous events.
- A short technical presentation delivered to the CTO and CEO summarising methodology, results, and lessons learned.

3. Dataset Description

A synthetic but realistic two-year sensor dataset has been pre-generated for this project. The file sensor_data_2yr.csv covers January 1, 2024 to December 31, 2025 at five-minute granularity, yielding 210,528 rows.

3.1 Sensor Channels

Column	Unit	Normal Range	Characteristics
temperature_c	°C	7 – 32	Daily + yearly sinusoidal cycles, Gaussian noise
vibration_mm_s	mm/s	1.5 – 5.5	Higher during business hours (08:00–18:00), exponential noise
pressure_kpa	kPa	100.8 – 101.8	Gentle yearly trend, Gaussian noise

Column	Unit	Normal Range	Characteristics
current_a	A	4.2 – 8.8	Bimodal (off-hours vs business hours)
anomaly	0 / 1	0 (normal)	Ground-truth label for model evaluation

3.2 Injected Anomaly Types

Approximately 1.09% of rows (~2,292) are labelled as anomalies. Three distinct fault patterns are present:

- Point spikes: sudden large increases in temperature, vibration, or current lasting one to a few readings.
- Pressure dropouts: sudden pressure drops of 5–10 kPa simulating sensor disconnection or valve failure.
- Gradual drift: a 500-step linear temperature ramp (~41 hours) simulating slow sensor degradation.

The anomaly column provides ground-truth labels so interns can measure precision, recall, and F1 score for any model they train.

4. System Architecture

The project is divided into three loosely coupled layers that map directly to each intern's primary ownership:

Layer	Owner (Primary)	Responsibilities
ML Pipeline	Sudarshini / Srivatsa	EDA, feature engineering, model training (Isolation Forest or Autoencoder), evaluation, model serialisation (joblib/ONNX)
REST API	Srivatsa / Sudarshini	FastAPI app with /predict and /stats endpoints, loads serialised model, returns JSON
Frontend Dashboard	Arshita	Chart.js or Recharts time-series plot, anomaly highlighting, summary panel, calls backend API

All three layers communicate through a well-defined JSON API contract agreed upon at the start of Week 1. This allows the frontend and ML tracks to develop in parallel from Week 2 onward.

5. Weekly Milestones

Week 1 (June 8 – 14) — Setup & Exploration

All three interns together:

- Set up shared GitHub repository, Python virtual environment, and Node environment.

- Explore the dataset: plot each sensor channel, compute descriptive statistics, visualise the anomaly distribution.
- Agree on the API contract: request/response JSON schema for /predict and /stats endpoints.
- Deliverable: a short EDA notebook or slide summarising dataset findings, pushed to GitHub.

Week 2 (June 15 – 21) — Model Training & API Skeleton

ML track (Sudarshini & Srivatsa):

- Engineer features: rolling mean, rolling standard deviation, z-score per channel.
- Train baseline Isolation Forest; evaluate with precision, recall, F1 against the anomaly label.
- Optional stretch: try a simple LSTM Autoencoder and compare reconstruction error approach.
- Serialise the best model.

Frontend track (Arshita):

- Scaffold React app (or vanilla HTML/JS with Chart.js).
- Render static time-series chart from a local JSON sample.
- Build summary panel layout (anomaly count, severity breakdown).

Week 3 (June 22 – 28) — Integration

- Srivatsa wires the serialised model into a FastAPI app; expose /predict (POST, accepts JSON rows) and /stats (GET, returns aggregate counts).
- Arshita connects the frontend to the live API; replace static sample data with real API responses.
- End-to-end smoke test: full dataset flows from CSV through API to rendered chart.
- Deliverable: working demo accessible on localhost, video screen-recorded as backup.

Week 4 (June 29 – July 4) — Polish & Presentation

- Fix integration bugs, improve UI clarity, add loading states and error handling.
- Write a one-page technical summary (methodology, results table, key learnings).
- Prepare and rehearse a 10-minute demo presentation for the CTO and CEO.
- Final deliverable: GitHub repo with README, trained model artefact, and recorded demo.

6. Recommended Tech Stack

Component	Recommended Tools
Data exploration	Python, pandas, matplotlib / seaborn, Jupyter Notebook
ML modelling	scikit-learn (Isolation Forest), optionally TensorFlow/Keras (Autoencoder)

Component	Recommended Tools
API layer	FastAPI, uvicorn, joblib (model loading)
Frontend	React (create-react-app) + Recharts, or HTML/JS + Chart.js
Version control	Git + GitHub (shared private repo)
Communication	Slack channel #intern-2026, weekly check-in with mentor

7. Evaluation Criteria

The final project will be assessed across four dimensions:

Criterion	Weight	Description
Model performance	30%	F1 score on anomaly label; correct identification of all three fault types
Code quality & collaboration	25%	Clean, commented code; meaningful Git commits; evidence of code review
Dashboard usability	25%	Intuitive UI, clear anomaly highlighting, responsive to API data
Presentation	20%	Clarity, technical depth, ability to answer questions from the mentor

8. Expected Learning Outcomes

- Hands-on experience with the end-to-end data science lifecycle on a real-world-style dataset.
- Practical understanding of anomaly detection algorithms and time-series feature engineering.
- Experience building and consuming REST APIs in a multi-person project.
- Exposure to frontend data visualisation with live data feeds.
- Collaboration skills: Git workflow, API contracting, and async communication across tracks.
- Domain knowledge in IoT sensor data patterns directly relevant to Infiswift's products.

9. Appendix — Dataset Quick Reference

File: sensor_data_2yr.csv

Property	Value
Date range	2024-01-01 00:00 – 2025-12-31 23:55

Property	Value
Granularity	5 minutes
Total rows	210,528
Total columns	6 (timestamp + 4 sensors + anomaly label)
Anomaly rows	~2,292 (~1.09%)
Anomaly types	Temperature spikes, vibration spikes, current spikes, pressure dropouts, gradual temperature drift
File size (approx)	~18 MB uncompressed

Suggested first commands:

```
import pandas as pd
df = pd.read_csv('sensor_data_2yr.csv', parse_dates=['timestamp'])
df.set_index('timestamp', inplace=True)
print(df.describe())
df[df.anomaly == 1].groupby(df[df.anomaly==1].index.date).size()
```